

VC dimension puzzle - medium

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1 Preliminaries

We consider a family $\mathcal{H} \subseteq 2^{[n]}$, identified with a family of Boolean vectors in $\{0,1\}^n$.

For $h \in \{0,1\}^n$ and $S \subseteq [n]$, we write $h|_S$ for the restriction of h to the coordinates in S . The Hamming weight of h is

$$|h| := |\{i \in [n] : h_i = 1\}|.$$

Definition 1 (VC dimension). *A set $S \subseteq [n]$ is shattered by $\mathcal{H}$ if*

$$\{h|_S : h \in \mathcal{H}\} = \{0,1\}^S.$$

The VC dimension of $\mathcal{H}$, denoted $\text{VCdim}(\mathcal{H})$, is the maximum cardinality of a set $S \subseteq [n]$ shattered by $\mathcal{H}$.

2 Problem formulation

Let $n \geq 3$ and let $\mathcal{H} \subseteq 2^{[n]}$ satisfy the following two properties:

- For every three distinct indices $i, j, k \in [n]$, there is $h \in \mathcal{H}$ such that

$$h|_{\{i,j,k\}} = 111.$$

- $\text{VCdim}(\mathcal{H}) \leq 2$.

Then $\mathcal{H}$ contains a vector of Hamming weight strictly greater than

$$\frac{n}{2}.$$